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Natural Language Processing System-

Recurrence Analysis

Given:

$$T(n) = T\left[\frac{n}{2}\right] + n$$

we will solve this using the substitution / Expansion Method.

① Expand the Recurrence

Starts with:

$$T(n) = T\left[\frac{n}{2}\right] + n$$

substitute $T[n/2]$:

$$T(n/2) = T(n/4) + n/2$$

Therefore,

$$T(n) = T(n/4) + n/2 + n$$

Expand Again:

$$T(n/4) = T(n/8) + n/4$$

so, $T(n) = T(n/8) + n/4 + n/2 + n$

Continuing:

$$T(n) = T\left[n/2^k\right] + n + n/2 + n/4 + \cdots + n/2^{k-1}$$

2. Find when Recursion stops

The recursion reaches the base case when:

$$\frac{n}{2^k} = 1$$

Therefore: $n = 2^k$

Taking log: $k = \log_2 n$

So there are approximately $\log_2 n$ levels.

3. Simplify the summation

we have:

$$T(n) = T(1) + n + \frac{n}{2} + \frac{n}{4} + \cdots + \frac{n}{2^{\log_2 n - 1}}$$

The summation is a Geometric series

$$n\left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots\right]$$

Using $1 + \frac{1}{2} + \frac{1}{4} + \cdots < 2$

we Get $T(n) < T(1) + 2n$

$\therefore$ $\boxed{T(n) = \Theta(n)}$

it is also:

$O(n)$

and $\Omega(n)$

Therefore, NLP System has Linear time Complexity

4. what does this Mean for an NLP System

Text Corpus Size	Approximate work
1,000 words	$O(1,000)$
10,000 words	$O(10,000)$
100,000 words	$O(100,000)$
1,000,000 words	$O(1,000,000)$

Because the Complexity is Linear, if the Corpus size becomes 10 times Larger, the processing work is Approximately 10 Times Larger.

For Example

$$n \to 2n$$

means Approximately:

$$T(n) \to 2T(n)$$

So performance decreases predictables as the copus Grows, rather than Exploding Exponentially

⑤ Efficiency Interpretation

The recurrence Contains two important operations:

* The problem size is reduced by Half ect Every recursive step: $n \to n/2$

* But the work done at Each Level is proportional to the Current input size.

Thus: $n + \frac{n}{2} + \frac{n}{4} + \frac{n}{8} + \cdots < 2n$

Even though there are logn recursive levels, the total work is done is dominated at the first level.

Therefore:

$$\boxed{\text{Time Complexity} = \Theta(n)}$$

Interpretation:

The NLP system is reasonably Efficient for Increasing Corpus sizes because its Running time Grows Linearly with the Amount of Text. However, very Large Corpora will still require proportionally more processing time and Computational resources.